

Qikang Huang

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EDUCATION

University of Michigan, Ann Arbor

Sep. 2026 – May 2028

M.S. in Electrical and Computer Engineering

Track: Signal & Image Processing and Machine Learning

- **Core Courses:** ECE 551 - Matrix Methods for Signal Processing, ECE 516 / BIOMEDE 516 — Medical Imaging Systems

Communication University of China / Coventry University

Sep. 2022 – Jun. 2026

Sino-foreign Cooperative Education Program

B.Eng. in Intelligent Science and Technology (Communication University of China)

BSc (Hons) in Computer Science with Artificial Intelligence, First Class Honours (Coventry University)

- **GPA:** 88/100
- **Core Courses:** Large Models and Deep Learning (98), Neural Networks and Deep Learning (97), Machine Learning (92), Computer Vision (91)

PROFESSIONAL EXPERIENCE

iFLYTEK Co., Ltd., Zhengzhou, China

Jan. 2026 – Apr. 2026

Assistant Research Algorithm Engineer – Smart Livestock Vision System

- Contributed to YOLOv8-based pig vision models for smart farming scenarios, including pig counting, posture detection, backfat-related classification, and abnormal-case monitoring.
- Refined the training-data feedback loop by validating 10,000+ RGB/depth images in LabelMe and correcting missed, false, duplicate, and inaccurate bounding-box annotations.
- Analyzed production failure cases from inspection robots and pig-counting modules, categorized false positives and false negatives, and supported model retraining and iteration.

ByteDance, Beijing, China

Jul. 2025 – Sept. 2025

Algorithm Intern – Loan Default Prediction Model Development and Optimization

- Built an end-to-end credit-risk pipeline for loan-default prediction, covering data cleaning, feature engineering, model training, hyperparameter tuning, and evaluation.
- Engineered time, categorical, binned, correlation, time-window, and group-ratio features, and compared LightGBM, XGBoost, and CatBoost using stratified K-fold cross-validation and ROC-AUC.
- Applied PSI-based stability selection, L2 regularization, early stopping, and stacking; the optimized XGBoost model achieved an AUC of approximately 0.738.

Business Intelligence of Oriental Nations Corporation Ltd., Beijing, China

Jan. 2025 – Feb. 2025

Data Analyst Assistant – Telecom Traffic Revenue Forecasting

- Supported a telecom operator mobile data-traffic revenue forecasting project to predict future revenue and identify key business drivers.
- Analyzed traffic-revenue time series with ACF/PACF, built ARIMA, SARIMA, ETS, and Prophet models, and evaluated performance using MAE.
- Selected SARIMA as the best-performing model, completed a 14-month revenue forecast, and proposed optimization recommendations on traffic unit price, average usage, billed users, and zero-billed users.

RESEARCH EXPERIENCE

Retrieval Strategy Optimization for Multi-turn Legal Consultation Systems Based on RAG

Jan. 2026 – May 2026

Undergraduate Thesis

- Designed a RAG-based legal retrieval framework for multi-turn consultations, addressing context dependence, omitted details, colloquial queries, and formal-language mismatch.
- Built a LexRAG pipeline over 17,228 legal articles and 5,065 multi-turn queries, evaluating recall and ranking quality with Recall@K and NDCG@K.
- Integrated query rewriting, BGE-M3 hybrid retrieval, HyDE query enhancement, and Cross-Encoder reranking to improve candidate article recall and ranking performance.
- The HyDE-enhanced hybrid retrieval strategy improved Recall@10 from 0.3757 to 0.4701 and NDCG@10 from 0.2651 to 0.3349, providing empirical support for retrieval-chain optimization in legal RAG systems.
- Found that key bottlenecks were concentrated in query representation modeling and legal terminology expansion rather than simple reranking of candidate results.

SKILLS & INTERESTS

- **Technical Skills:** AI-Assisted Development, Python, C/C++, PyTorch, NumPy, OpenCV, Matplotlib, LaTeX
- **Languages:** English (Proficient), Chinese (Native)
- **Interests:** Fitness, Hiking, Swimming